## Question 1 2 marks 101

A central angle of a circle subtends an arc length of $5\pi$ cm.
Given the circle has a radius of 9 cm, find the measure of the central angle in degrees.

## Question 2 4 marks 102

Solve the equation $\csc^2\theta + 3\csc\theta - 4 = 0$ over the interval $[0, 2\pi]$.
Express your answers as exact values or correct to 3 decimal places.

## Question 3 3 marks 103

Jess invests \$12 000 at a rate of 4.75% compounded monthly.
How long will it take for Jess to triple her investment?

Express your answer in years, correct to 3 decimal places.

## Question 4 3 marks 104

The 4th term in the binomial expansion of $\left(qx^2 - \dfrac{3}{x}\right)^{10}$ is $414\ 720x^{11}$.

Determine the value of $q$ algebraically.

**Note: A calculator is not required for the remaining test questions.**

## Question 5 1 mark 105

Bella has 2 pairs of shoes, 3 pairs of pants, and 10 shirts.
Carey has 4 pairs of shoes, 4 pairs of pants, and 4 shirts.
An outfit is made up of one pair of shoes, one pair of pants, and one shirt.

Who can make more outfits? Justify your answer.

## Question 6 2 marks 106

In the binomial expansion of $(x - y)^{10}$, how many terms will be positive?

Justify your answer.

## Question 7 4 marks

107

Solve the following equation algebraically where $180^\circ \leq \theta \leq 360^\circ$.

$$2\sin^2\theta + 5\cos\theta + 1 = 0$$

## Question 8 3 marks 108

Solve the following equation algebraically:

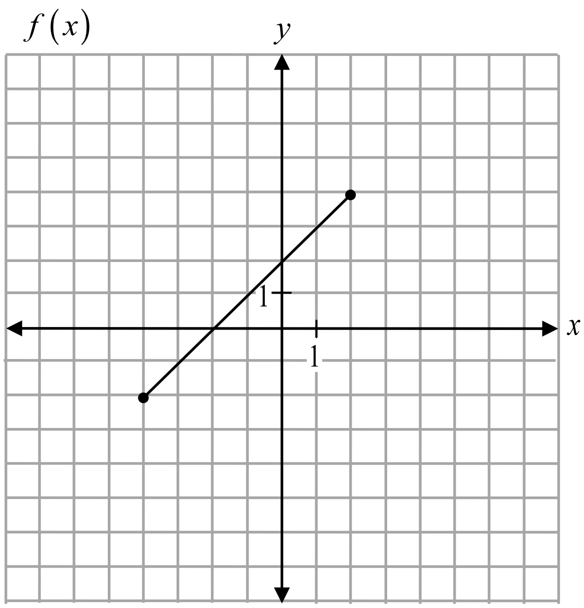
$$\log_3(x - 4) + \log_3(x - 2) = 1$$

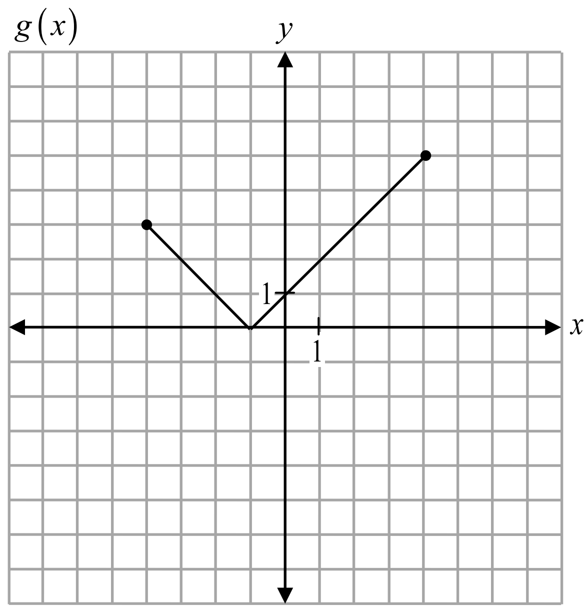
## Question 9 1 mark 109

Given that $f(x) = \{(1, 3), (2, 5), (3, 4), (4, 2)\}$, find $f(f(3))$.

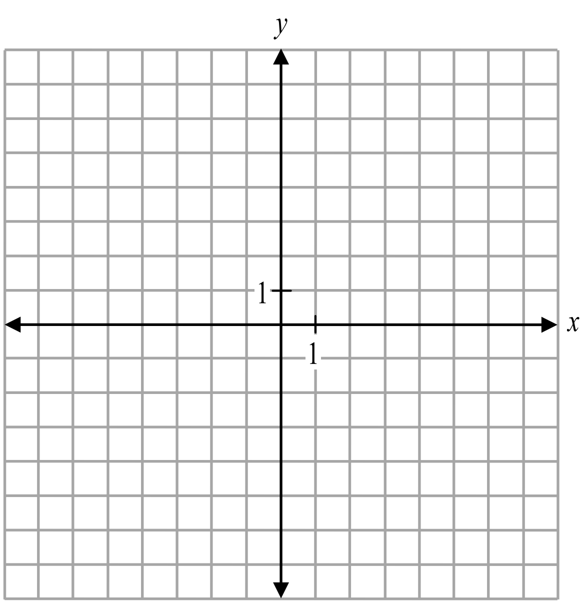
## Question 10 2 marks 110

Given the graphs of $f(x)$ and $g(x)$ below,




sketch the graph of $y = f(x) - g(x)$.



## Question 11 2 marks 111

Given the graph of $y = f(x)$, describe the transformations to obtain the graph of the function $y = f(2x - 6)$.

## Question 12 1 mark 112

Given $f(x) = \{(-3, 4), (2, 7), (8, 6)\}$, state the domain of the resulting function after $f(x)$ is reflected through the line $y = x$.

## Question 13 3 marks 113

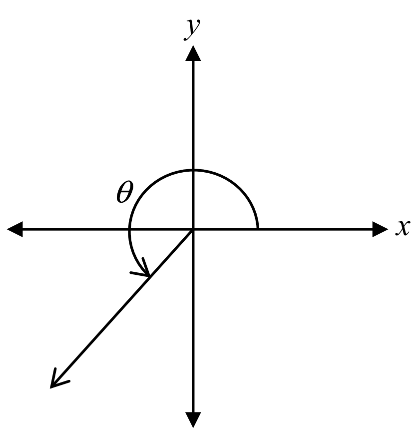
Determine the value of $y$ in the following equation:

$$\log_x 27 - \log_x 3 = 2\log_x y$$

## Question 14 1 mark 114

Angle $\theta$, measuring $\frac{5\pi}{4}$, is drawn in standard position as shown below.

Determine the measures of all angles in the interval $[-4\pi, 2\pi]$ that are coterminal with $\theta$.



## Question 15 — 3 marks 115

Prove the identity below for all permissible values of $x$:

$$\frac{\sin^2 x}{\sec x + 1} = \cos x - \cos^2 x$$

| Left-Hand Side | Right-Hand Side |
| --- | --- |
| | |

## Question 16 3 marks

116

Solve algebraically:

$${}_{n}C_{2} = 4n + 5$$

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